

Complete Assignment Answers (Questions 1-50)

- Q1. A variable is a named memory location used to store data.
- Q2. Variables store values that can change during program execution.
- Q3. Variable name is identifier; value is stored data.
- Q4. b) total_marks
- Q5. C++ is case-sensitive.
- Q6. No, variable names cannot contain spaces.
- Q7. Yes.
- Q8. Compilation error in same scope.
- Q9. studentName, studentAge, studentMarks
- Q10. studentMarks because it is meaningful.
- Q11. A data type defines the kind of data stored.
- Q12. int
- Q13. float or double
- Q14. double has higher precision.
- Q15. char
- Q16. bool
- Q17. string
- Q18. 18
- Q19. 25=int, 3.14=float/double, 'A'=char, true=bool
- Q20. Decimal part is truncated.
- Q21. To take input from user.
- Q22. To display output.
- Q23. >>
- Q24. <<
- Q25. Inputs value into age.
- Q26. Displays value of age.
- Q27. Moves cursor to next line.
- Q28. cin >> marks;
- Q29. cout << "Welcome to C++";
- Q30. Programming
- Q31. A symbol that performs operations.
- Q32. +
- Q33. 1
- Q34. 0
- Q35. ==
- Q36. = assigns, == compares.

Q37. true

Q38. Logical AND

Q39. Logical OR

Q40. Logical NOT

Program Based Questions (41-50)

41. Hello World

Problem Statement: Print Hello World

Algorithm: Read input, process, display result.

```
#include <iostream>
using namespace std;
int main(){
    cout<<"Hello World";
    return 0;
}
```

42. Sum of Two Numbers

Problem Statement: Input two numbers and display sum

Algorithm: Read input, process, display result.

```
#include <iostream>
using namespace std;
int main(){
    int a,b;
    cin>>a>>b;
    cout<<"Sum = "<<a+b;
    return 0;
}
```

43. Area of Rectangle

Problem Statement: Calculate area

Algorithm: Read input, process, display result.

```
#include <iostream>
using namespace std;
int main(){
    int l,b;
    cin>>l>>b;
    cout<<"Area = "<<l*b;
    return 0;
}
```

44. Formula

Problem Statement: Simple Interest = $(P \times R \times T) / 100$

Algorithm: Read input, process, display result.

45. Simple Interest Program

Problem Statement: Calculate SI

Algorithm: Read input, process, display result.

```
#include <iostream>
using namespace std;
int main(){
    float p,r,t;
    cin>>p>>r>>t;
    cout<<(p*r*t)/100;
    return 0;
}
```

46. Temporary Variable

Problem Statement: Used while swapping

Algorithm: Read input, process, display result.
temp=a; a=b; b=temp;

47. Swap Numbers

Problem Statement: Swap a and b

Algorithm: Read input, process, display result.
int a=10, b=20, temp;
temp=a;
a=b;
b=temp;

48. Formula

Problem Statement: $F = (C * 9/5) + 32$

Algorithm: Read input, process, display result.

49. Fahrenheit Value

Problem Statement: $0^{\circ}\text{C} = 32^{\circ}\text{F}$

Algorithm: Read input, process, display result.

50. Integer Division

Problem Statement: 9/5 gives 1; use 9.0/5

Algorithm: Read input, process, display result.